

pocket_sdr.py help

ver. 0.17 2026-06-17

Initial Window

Page Tabs:
Tabs to switch current page

Load or Save Receiver Options:
The options format complies with Windows ini.

Status Bar:
Status info or error message for the receiver

Command Buttons

Start	: Start the receiver
Stop	: Stop the receiver
Input...	: Show Input Options dialog
Output...	: Show Output Options dialog
Signal...	: Show Signal Options dialog
System...	: Show System Options dialog
Array...	: Show Array Options dialog
Help	: Show Help dialog
Exit	: Exit the receiver

Receiver Time:
Elapsed time since the receiver started (s)

Receiver Page

Receiver Time:

Elapsed time since the receiver started

Input Source:

IF data source (RF_Frontend or IF_Data)

Fmt/# RF CH/# Array CH:

IF data format *3 and number of CHs

LO Frequencies (MHz):

Local oscillator frequencies for each RF CH (MHz)

Sampling:

IF data sampling type for each RF CH (I or IQ)

Sampling Rate (Mps):

IF data sampling rate

BB CH Locked/All:

Number of BB CHs tracked and all

IF Data Rate (MB/s):

IF data transfer rate

IF Data Buffer Usage (%):

Receiver internal IF data buffer usage rate

Time (GPST):

PVT Solution time in GPST

Solution Status:

PVT Solution status (- or FIX)

Latitude, Longitude and Altitude:

PVT Solution latitude/ longitude (deg) and ellipsoidal height (m)

Roll/Pitch/Yaw (deg):

Attitude solution by antenna array

Sats Used / All:

Number of used satellites for PVT and all satellites with OBS data

Solution Latency (s):

PVT Solution latency

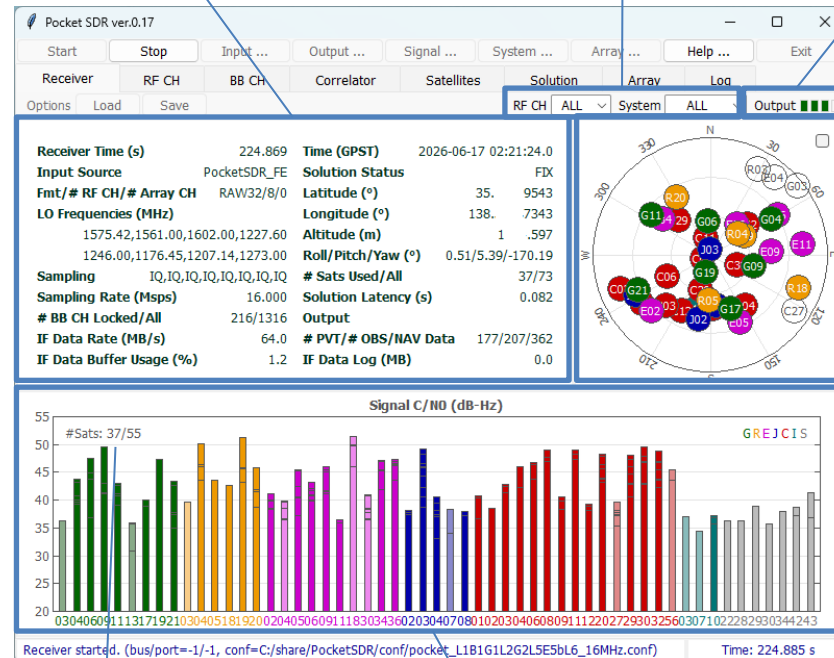
PVT/# OBS/NAV Data:

Total count of PVT solutions, OBS and NAV data

IF Data Log (MB):

Recorded size of IF Data log

Receiver Status Panel



RF CH and System Selection for Skyplot and C/N0 Plot

Output Status Indicators

for PVT solution, OBS/NAV data, receiver log and IF data log

Orange: waiting for connection

Green: data OK

Light-Green: data active

Red: error

White: no output

Satellite Positions in Skyplot *1 *2

Number of used satellites and all satellites with OBS data

Signal C/N0 (dB-Hz) Plot for Each Satellite and Signals (stacked) *1 *2

* 1 GNSS System Color:

Green: GPS

Orange: GLONASS

Magenta: Galileo

Blue: QZSS

Red: BeiDou

Blue-Green: NavIC

Grey: SBAS

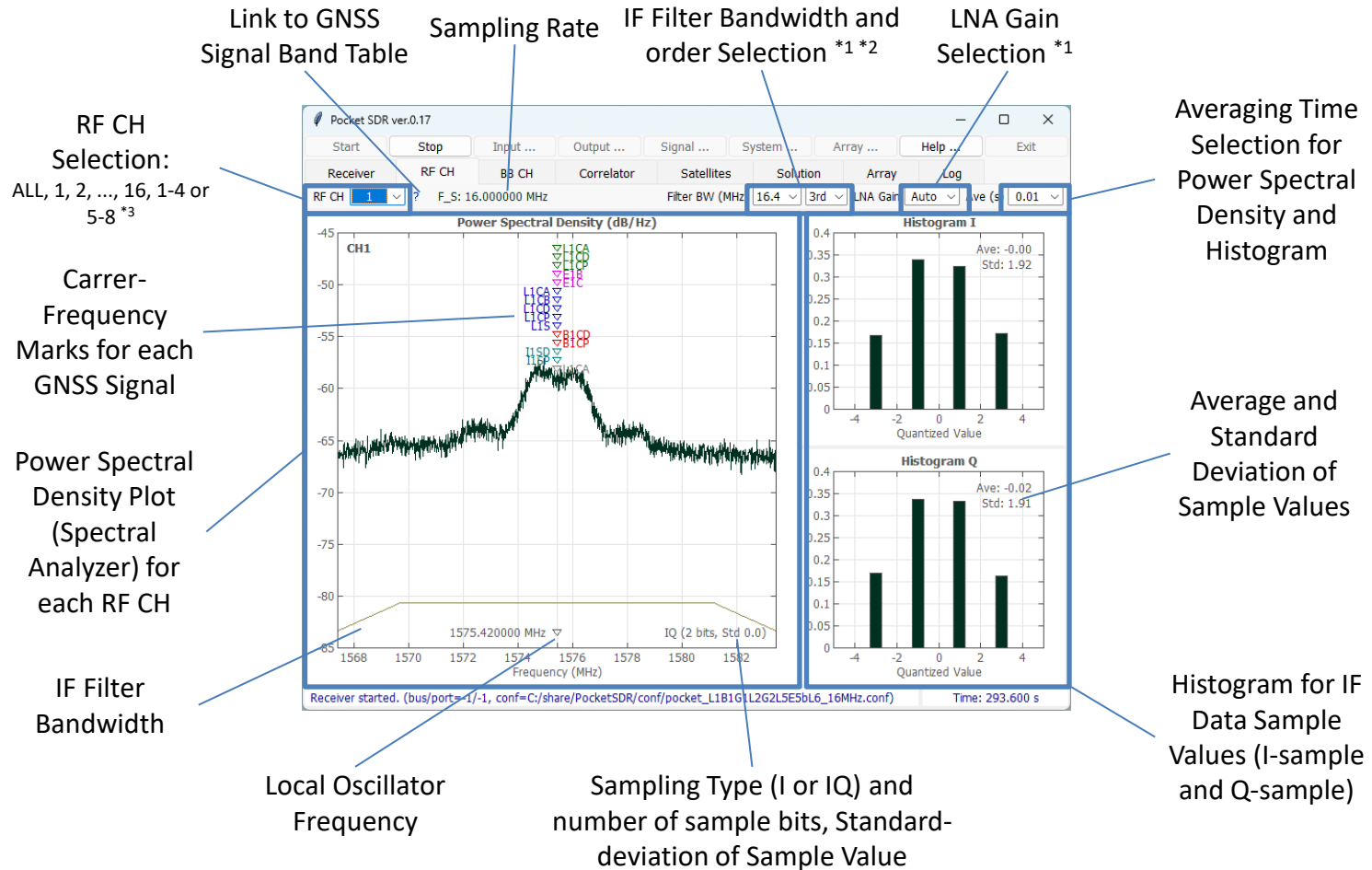
White or light-color:

unused satellites for solution

*2 Satellite ID complies with RINEX convention

*3 INT8 (int8 I-sampling), INT8X2 (interleaved-int8 IQ-sampling), RAW8, RAW16, RAW32 (Pocket SDR FE), CS8 or CS16 (SoapySDR devices)

RF CH Page



*1 Only valid for RF_Frontend Input, *2 Only valid for IQ-sampling

*3 RF CH 9, 10, ..., 16 are beamforming CHs by antenna array

BB CH Page

RF CH Selection:

ALL, 1, 2, ..., 16

GNSS System Selection:

ALL, GPS, GLONASS, ...

BB CH State Selection:

LOCK or ALL

Link to GNSS Signal Table

Internal IF Data Buffer Usage Rate in %

BB CH No Searching Signal

Status for each BB CH:

CH : BB CH no

RF : Assigned RF CH no for BB CH

SAT : Satellite ID

SIG : Signal ID

PRN : PRN number or FCN for GLONASS FDMA signal

LOCK : Continuous lock time (s)

C/N0 : Signal C/N0 (dB-Hz)

(dB-Hz) : Signal C/N0 Bar (20-50 dB-Hz)

COFF : Code offset (ms)

DOP : Doppler frequency (Hz)

ADR : Accumulated Doppler range (cycle)

SYNC : Synchronization status

#NAV : Decoded subframe or message count

#ERR : NAV data error count

#LOL : Loss-of-lock count

FEC : Number of error bits corrected by FEC (-1: correction unable)

Number of Locked and All BB CHs

Synchronization Status:

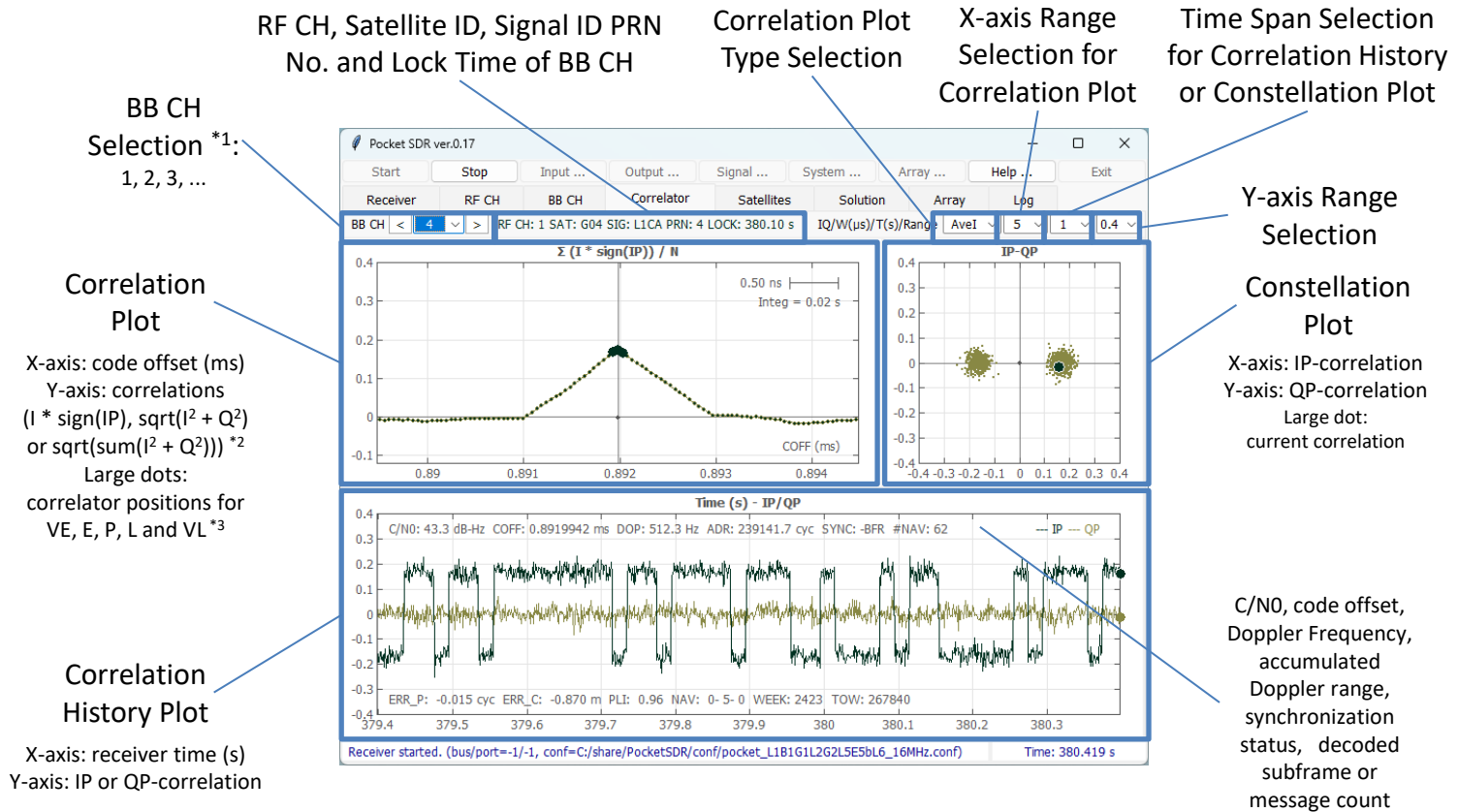
S : Secondary code sync

B : NAV symbol (bit) sync

F : Subframe or message sync

R : Polarity of subframe or message sync (R: reversed)

Correlator Page



*1 BB CH can be selected by clicking BB CH Status row in BB CH Page

*2 Integration time is based on non-coherent integration time for DLL

*3 VE and VL are only shown in case of BOC-modulation and Bump-Jump enabled

Satellites Page

System Selection

Status for each Satellite:

SAT : Satellite ID
 FCN : Frequency channel no (FCN) for GLONASS FDMA signals
 PVT : Satellite used in PVT solution or not
 OBS : Satellite L1 OBS data availability
 EPH : Satellite ephemeris availability
 SVH : Satellite health flag in ephemeris (HEX)
 AZ : Azimuth angle (deg)
 EL : Elevation angle (deg)
 SIG1 : Signal 1 ID
 C/N0 : C/N0 of signal 1 (dB-Hz)
 SIG2 : Signal 2 ID
 C/N0 : C/N0 for signal 2 (dB-Hz)
 SIG3 : Signal 3 ID
 C/N0 : C/N0 for signal 3 (dB-Hz)
 ...

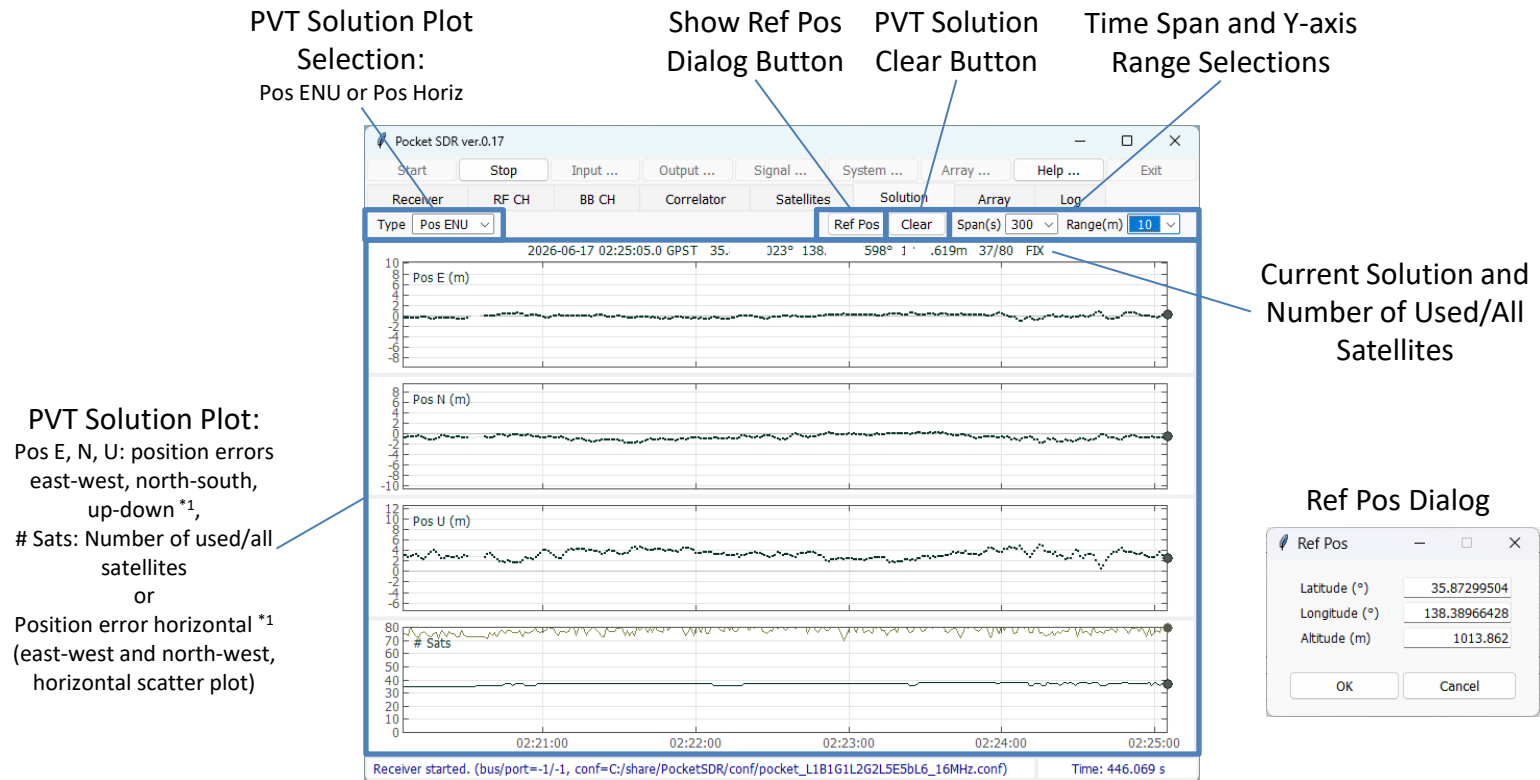
Number of satellites used for PVT solution, number of satellites tracked, number of all tracked signals

Red: unhealthy satellite
 Grey: unused satellite for PVT solution

SAT	FCN	PVT	OBS	EPH	SVH	AZ(°)	EL(°)	SIG1	C/N0	SIG2	C/N0	SIG3	C/N0	SIG4	C/N0	SIG5	C/N0	SIG6	C/N0
G04	-	OK	OK	OK	00	58.0	35.2	L1CA	42.9	L1CD	40.1	L1CP	44.4	L2CM	39.5	L5I	41.2	L5Q	40.4
G06	-	OK	OK	OK	00	359.1	64.8	L1CA	48.5	L2CM	44.3	L5I	46.0	L5Q	37.3	-	-	-	-
G09	-	OK	OK	OK	00	101.0	55.8	L1CA	47.7	L2CM	39.8	L5I	44.1	L5Q	42.0	-	-	-	-
G11	-	OK	OK	OK	00	305.1	36.6	L1CA	45.7	L1CD	41.5	L1CP	46.2	L2CM	41.3	L5I	41.9	L5Q	44.0
G17	-	OK	OK	OK	00	159.2	44.7	L1CA	42.6	L2CM	36.3	-	-	-	-	-	-	-	-
G19	-	OK	OK	OK	00	192.8	74.7	L1CA	46.4	-	-	-	-	-	-	-	-	-	-
G21	-	OK	OK	OK	00	245.1	29.0	L1CA	42.6	L1CD	38.7	L1CP	43.6	L2CM	36.2	L5I	34.5	L5Q	36.2
R03	+5	-	OK	OK	00	28.8	13.4	G1CA	37.5	-	-	-	-	-	-	-	-	-	-
R04	+6	OK	OK	OK	00	48.8	62.2	G1CA	50.1	G2CA	47.5	G3OCD	42.7	G3OCP	42.4	-	-	-	-
R05	+1	OK	OK	OK	00	182.2	57.1	G1CA	44.0	G3OCD	34.4	G3OCP	34.2	-	-	-	-	-	-
R18	-3	OK	OK	OK	00	111.7	17.0	G1CA	46.7	-	-	-	-	-	-	-	-	-	-
R19	+3	OK	OK	OK	00	64.7	59.7	G1CA	51.8	G2CA	46.5	G3OCD	42.2	G3OCP	42.1	-	-	-	-
R20	+2	OK	OK	OK	00	330.9	39.9	G1CA	46.6	G2CA	43.0	G3OCD	37.3	G3OCP	37.0	-	-	-	-
E02	-	OK	OK	OK	00	225.4	26.7	E1B	35.3	E1C	35.0	ESAI	38.1	ESAQ	38.1	ESBI	37.7	ESBQ	37.6
E04	-	-	OK	OK	00	38.9	12.4	E1B	35.0	E1C	35.0	ESBI	35.2	ESBQ	35.3	E6B	36.4	E6C	36.5
E05	-	OK	OK	OK	00	155.6	34.9	E1B	40.2	E1C	40.0	ESAI	43.1	ESAQ	43.1	ESBI	41.9	ESBQ	42.1
E06	-	OK	OK	OK	00	57.8	29.8	E1B	41.3	E1C	41.5	ESAI	40.5	ESAQ	40.3	ESBI	39.3	ESBQ	39.3
E09	-	OK	OK	OK	00	85.1	43.3	E1B	45.1	E1C	45.8	ESAI	43.0	ESAQ	43.1	ESBI	41.4	ESBQ	41.4
E11	-	OK	OK	OK	00	84.5	18.3	E1B	32.4	E1C	32.5	ESBI	34.9	ESBQ	34.8	-	-	-	-
E18	-	OK	OK	92	0.0	0.0	E1B	50.5	E1C	49.7	ESAI	47.0	ESAQ	46.9	ESBI	47.6	ESBQ	47.3	
E30	-	OK	OK	OK	00	284.2	26.9	E1B	38.9	E1C	38.6	ESAI	37.3	ESAQ	37.3	ESBI	36.8	ESBQ	36.7
E34	-	OK	OK	OK	00	308.3	44.7	E1B	45.4	E1C	45.7	ESAI	42.9	ESAQ	43.1	ESBI	41.9	ESBQ	41.6
E36	-	OK	OK	OK	00	42.7	58.4	E1B	45.8	E1C	45.7	ESAI	45.1	ESAQ	45.0	ESBI	43.9	ESBQ	43.8
J02	-	OK	OK	OK	00	191.1	39.4	L1CA	38.0	L1CP	38.7	L1S	37.9	-	-	-	-	-	-
J03	-	OK	OK	OK	00	355.2	85.2	L1CA	46.4	L1CD	43.7	L1CP	48.1	L1S	48.5	L2CM	44.7	L5I	38.6
J04	-	OK	OK	OK	10	173.9	51.3	L1CB	37.3	L1CD	33.0	L1CP	38.0	L1S	38.1	L2CM	35.6	L5I	37.0
J07	-	-	OK	OK	00	0.0	0.0	L1CP	34.5	L1S	39.9	L2CM	34.1	L6E	34.3	-	-	-	-
J08	-	OK	OK	OK	10	242.1	25.0	L1CB	36.6	L1CP	36.4	L5Q	36.1	L5SQV	36.4	L6E	34.2	-	-
C01	-	OK	OK	OK	00	177.2	48.9	B1I	34.7	B2BI	35.3	B3I	34.1	-	-	-	-	-	-
C02	-	OK	OK	OK	00	249.1	16.0	B1I	36.0	-	-	-	-	-	-	-	-	-	-

Receiver started. (bus/port=-1/-1, conf=C:/share/PocketSDR/conf/pocket_L1B1G1L2G2L5E5bL6_16MHz.conf) Time: 408.091 s

Solution Page



*¹ The reference position is automatically selected from the first epoch solution.
It can be modified by using Ref Pos Dialog

Array Page

Array Calibration Status:
STOP or RUN, Total epochs for calibration and RMS residuals

Array Calibration Targets:
Both: Dealy and Attiude
Delay: Delay only
Att: Attitude only

Array Calibration Operation Buttons:
Start or Stop, Clear, Load and Save

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Start Stop Input ... Output ... Signal ... System ... Array ... Help ... Exit

Receiver RF CH BB CH Correlator Satellites Solution Array Log

CALIB: STOP EPOCHS: 0 RMS: 0.0000 m

Calibration Both Start Clear Load Save

RF CH DELAY

CH1: 0.0000 m CH2: 0.0044 m CH3:-0.0039 m CH4: 0.0787 m
CH5:-0.0268 m CH6:-0.0519 m CH7:-0.0422 m CH8: 0.0000 m

ARRAY ATTITUDE

ROLL: 0.511° PITCH: 5.392° YAW:-170.194°

ARRAY CH BEAM DIRECTION

CH	AZ	EL	Update
CH9	0.000 °	90.000 °	Update
CH10	0.000 °	90.000 °	Update
CH11	0.000 °	90.000 °	Update
CH12	0.000 °	90.000 °	Update
CH13	0.000 °	90.000 °	Update
CH14	0.000 °	90.000 °	Update
CH15	0.000 °	90.000 °	Update
CH16	0.000 °	90.000 °	Update

Receiver started. (bus/port=-1/-1, conf=C:/share/PocketSDR/conf/pocket_L1B1G1L2G2L5E5bL6_16MHz.conf) Time: 542.589 s

RF CH DELAY:
RF CH delay calibration results referenced to RF CH1

ARRAY ATTITUDE:
Array attitude calibration results (ROLL, PITCH, YAW angles)

ARRAY CH BEAM DIRECTION:
Beam direction control for each Array CH. Set AZ (azimuth) and EL (elevation) angles and push Update Button.

Log Page

Additional Receiver Log Filter:
filter string, like G01 or L1CA, separated
by spaces, | indicates OR condition
(e.g., G10|G02|G03)

Receiver Log
Type Selection:
Blank: all log

Receiver
Logs *1 *2

Receiver Log
Pause Button

Receiver Log
Clear Button

SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C04, 1, 1, 40.0, 145.8, 42.8, -2.668
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C06, 1, 1, 46.2, 246.5, 53.1, -0.383
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C08, 1, 1, 49.1, 237.1, 82.7, 0.324
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C09, 1, 1, 40.0, 236.7, 26.5, 2.950
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C11, 1, 1, 49.3, 348.9, 79.0, 0.180
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C12, 1, 1, 39.2, 208.2, 44.6, 1.395
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C19, 1, 1, 35.6, 212.6, 17.6, -1.203
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C20, 1, 1, 46.6, 197.0, 64.9, 0.007
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C27, 0, 1, 40.3, 125.1, 10.6, 0.000
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C29, 1, 1, 48.3, 317.7, 56.5, 0.303
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C30, 1, 1, 48.5, 116.1, 66.0, -0.505
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C32, 1, 1, 47.1, 47.9, 50.8, -0.214
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, C56, 0, 1, 46.0, 0.0, 0.0, 0.000
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, I10, 1, 1, 41.2, 194.9, 47.0, -0.000
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, S22, 0, 1, 35.4, 0.0, 0.0, 0.000
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, S28, 0, 1, 38.4, 0.0, 0.0, 0.000
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, S29, 0, 1, 40.0, 0.0, 0.0, 0.000
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, S34, 0, 1, 37.6, 0.0, 0.0, 0.000
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, S37, 0, 1, 38.2, 0.0, 0.0, 0.000
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, S42, 0, 1, 37.9, 0.0, 0.0, 0.000
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, S43, 0, 1, 44.4, 0.0, 0.0, 0.000
 SSAT, 591.920, 2026, 6, 17, 2, 27, 32.000, S44, 0, 1, 36.6, 0.0, 0.0, 0.000
 SNAV, 585.992, C11, B11, 11, 0, 300, E240D059C0010C5A135F7AAF917F65F31FF6E7061FF3B8C7B7C6A38CB32D617EE305C790CE80
 SNAV, 586.039, C08, B11, 8, 0, 300, E240D059C0010C5A138079DD21FFABE6BFFFE85FFB839C6A73C61A6D9A08A31160702122830
 SNAV, 586.040, C56, B11, 56, 0, 300, E240D059C0056C8F1254F60CAAFB9F63FFE7C32DFF2230FF0E1E33EF4DE2E810E2DEE5209C0
 SNAV, 586.042, C06, B11, 6, 0, 300, E240D059C0010C5A140EAD1C47FD38FDDFF873FD601074F79E6D2C0913BDF13DE5BACD6A160
 SNAV, 585.996, C12, B31, 12, 0, 300, E240D059C0010C5A13D48BEDACFF8DE7FFF5A659C00D93DFBE3A6E396161702073C8FBBC6010
 SNAV, 586.005, C19, B11, 19, 0, 300, E240D059C0010C5A142A2FC55800A215DFF5668E4009908FBDFAEE48777333D0E94FF5034530
 SNAV, 585.994, C20, B11, 20, 0, 300, E240D059C0010C5A1429EF71CD80381F5FF66E6D160044A77B35E6E4C487599C0A044411AC8C0
 SNAV, 586.050, C09, B21, 9, 0, 300, E240D059C0010C5A1569EEF8F57F2CF11FFB8C2CE01A7D8792D86C11EEED5E7483B0DC314580
 SNAV, 586.039, C08, B31, 8, 0, 300, E240D059C0010C5A138079DD21FFABE6BFFFE85FFB839C6A73C61A6D9A08A31160702122830
 SNAV, 586.050, C09, B11, 9, 0, 300, E240D059C0010C5A1569EEF8F57F2CF11FFB8C2CE01A7D8792D86C11EEED5E7483B0DC314580
 SNAV, 586.050, C09, B31, 9, 0, 300, E240D059C0010C5A1569EEF8F57F2CF11FFB8C2CE01A7D8792D86C11EEED5E7483B0DC314580
 SNAV, 586.042, C06, B31, 6, 0, 300, E240D059C0010C5A140EAD1C47FD38FDDFF873FD601074F79E6D2C0913BDF13DE5BACD6A160
 SNAV, 586.040, C56, B11, 56, 0, 300, E240D059C0056C8F1254F60CAAFB9F63FFE7C32DFF2230FF0E1E33EF4DE2E810E2DEE5209C0

Receiver started. (bus/port=-1/-1, conf=C:/share/PocketSDR/conf/pocket_L1B1G1L2G2L5E5B16_16MHz.conf) Time: 592.657 s

*1 Refer p.16 for the receiver log formats

*2 To copy the logs to Clip-board, push Pause button, select the logs and push Ctrl-C of the keyboard

Input Options Dialog

Input Source Selection:
RF Frontend or IF Data

Device Configuration File:
RF Frontend configuration settings for Pocket SDR FE (MAX2771 register) to be loaded at the receiver started. Check Device Configuration File and fill the file path. If unchecked, the current configuration for the device is used. Ignored for other devices.

IF Data File *1:
IF data file path, if IF Data selected as the input source. The IF data can be captured by pocket_dump or as IF data log of the receiver with tag files.

Sampling Rate and Device Options for Soapy SDR Devices:
-GAIN=gain (LNA gain),
-BW=bw (Filter Bandwidth),
Options are separated by spaces.

RF Frontend Type Selection:
Pocket SDR FE: Pocket SDR FE 2CH, 4CH, 8CH
USRP (uhd): USRP via Soapy SDR
LimeSDR (lime): LimeSDR via Soapy
LimeSDR (limesuiteng): LimeSDR via newer driver with Soapy
BladeRF (bladrf): bladeRF via Soapy
RTL-SDR (rtlsdr): RTL-SDR via Soapy
Plute-SDR (plutosdr): PlutoSDR via Soapy
Airsipy (airsipy): Airspy via Soapy

USB Bus/Port:
To select Pocket SDR FE in case of multiple devices connected. If blank, the device is automatically selected.

Time Offset and Scale:
Start time offset and time scale to playback the IF Data.

IF Data File Property *1:
IF Data Format, Sampling Rate, LO frequency, Sampling Type (I or IQ), Number of sample bits for each RF CH. If the tag file exist for the IF data file, these are automatically configured according to the tag file.

Input LPF Settings:
Set two-sided bandwidth (MHz) of LPF for each RF CH. 0 indicates disabled.

*1 Refer to p.15 for the IF data file formats

Output Options Dialog

PVT Solution Output
Path:

PVT Solution (NMEA 0183
GNRMC, GNGGA, GNGSA,
GxGSV) path *¹.

OBS and NAV Data
Output Path:

OBS and NAV Data (RTCM 3.4
MSM (MT1077, 1087, 1097, 1117,
1127, 1137 and 1107) with
extension and EPH (MT1019,
1020, 1045, 1046, 1044, 1042
and 1041) path *¹ *²

Receiver Log
Output Path:

Receiver Log Path *¹ *³

Receiver Log Type
Selections *³

Execute RINEX
Converter for RTM3
OBS and NAV data

IF Data Log Output
Path:

IF Data Log Path *¹ *⁴,
It is only valid for RF
Frontend selected as the
input source

*¹ The output path should be one of the followings.

- (1) <local path>: A local file path <local path>. It can contain time keywords replaced by the log start time. Optionally, it can include the file swapping order “::S=<tint>” by the specified interval <tint> (H).
- (2) <:port>: TCP server with the port number <port>. The TCP server waits for and accepts the external TCP connection at the port and output data
- (3) <addr>:<port>: TCP client with the address <addr> and the port number <port>. The TCP client connects the external TCP server and output data

*² The output RTCM 3.4 MSM (with extensions) and EPH can be converted to RINEX OBS and NAV by CONVBIN utility included in the Pocket SDR package

*³ Refer to p.16 for the receiver log format. The receiver log can be plotted by the Log Viewer pocket_plot.py in the Pocket SDR package

*⁴ Refer p.15 for the IF data log format. The output format is automatically selected according to the RF Frontend (RAW8, RAW16, RAW32, CS8 or CS16). The tag file is also generated.

Signal Options Dialog

GNSS System and Signal Selections:

System: GNSS System,
Satellite No: Satellite number or
number range (n-m) separated
by , *¹
GNSS Signals: Signal IDs to be
tracked

RF CH Assignment Options:

RF CH is automatically assigned for
signals. To assign signals to specific
RF CH(s) manually, use the option
as the format <sig>:<ch_list>
separated by spaces, where, <sig>
is signal ID and <ch_list> is RF CH
number list like 1,3,5-7.
Additionally, SRCH:<ch_list> can be
used to specify signal search CH(s)
to shorten the acquisition time.

Set or unset all systems and signals

Link to Detailed GNSS
Signal Table

The screenshot shows the 'Signal Options' dialog box. It has a table with three main columns: 'System', 'Satellite No', and 'GNSS Signals'. The 'GNSS Signals' column contains checkboxes for various signal IDs. Below the table, there is a text field for 'RF CH Assignments (<sig>:<ch>[<ch>],...[...])'. At the bottom, there are buttons for 'Set All', 'Unset All', 'OK', and 'Cancel'.

Pocket SDR Signal ID Table

System	Signal	Signal ID	System	Signal	Signal ID
GPS	L1C/A	L1CA	BeiDou	B1I	B1I
	L1C-D	L1CD		B1C-D	B1CD
	L1C-P	L1CP		B1C-P	B1CP
	L2C-M	L2CM		B2a-D	B2AD
	L5-I	L5I		B2a-P	B2AP
GLONASS	L5-Q	L5Q	NavIC	B2I	B2I
	L1C/A	G1CA		B2b-I	B2BI
	L10Cd	G10CD		B3I	B3I
	L10Cp	G10CP		L1-SPS-D	I1SD
	L2C/A	G2CA		L1-SPS-P	I1SP
Galileo	L20Cp	G20CP	SBAS	L5-SPS	I5S
	L30Cd	G30CD		S-SPS	ISS* ³
	L30Cp	G30CP		L1C/A	L1CA
	E1-B	E1B		L5-I	L5I
	E1-C	E1C		L5-Q	L5Q
QZSS	E5a-I	E5AI			
	E5a-Q	E5AQ			
	E5AltBOC	E5ABQ			
	E5b-I	E5BI			
	E5b-Q	E5BQ			
QZSS	E6-B	E6B			
	E6-C	E6C			
	L1C/A	L1CA			
	L1C/B	L1CB			
	L1C-D	L1CD			
	L1C-P	L1CP			
	L1S	L1S			
	L2C-M	L2CM			
	L5-I	L5I			
	L5Q	L5Q			
	L5S-I	L5SI			
	L5S-Q	L5SQ			
	L5S-Q	L5SQV* ²			
	L6D	L6D			
	L6E	L6E			

*¹ For GPS, Galileo, BeiDou, NavIC and SBAS, these satellite numbers are the same as the signal PRN numbers. For GLONASS, use the frequency channel number (FCN) for FDMA signals and satellite slot number for CDMA signals instead separated by /.
For QZSS, which has complicated satellite numbering, use SV IDs (1-5, 8, 9) or non-standard code numbers (6, 10) instead as the satellite numbers.

*² QZSS L5S verification mode signals

*³ Pocket SDR FE currently does not support S-band signals

System Options Dialog

PVT Options Selection:

Epoch Interval (s), Max Epoch Lag (s) and Elevation angle Mask (deg)

Signal Acquisition and Tracking Options Selection:

Correlation Spacing (chip), Non-coherent Integration Time for Acquisition (s), Non-coherent Integration Time for DLL (s), Loop filter bandwidth for DLL, PLL, FLL (wide-band) and FLL (narrow-band) (Hz), Max Doppler Frequency search range (Hz), C/N0 thresholds for signal-lock or signal-lost decision (dB-Hz), Carrier Lock Threshold as PLI (phase lock indicator), Lost Decision Count, Bump-Jump for BOC modulation enabled or disabled, Max code length for direct acquisition (ms).

Signal Acquisition Mode Selection:

Full-Search: All PRN signals are searched.
Fast-Search: Only signals of visible satellites are searched for fast acquisition. The visibility is estimated with CPU time, last or current FIX position, almanac or ephemeris of each satellites. The Doppler also is decided by last or current reference oscillator freq. offset. So, if FE changed, use Full-Search.

FFTW Wisdom Path:

FFTW wisdom file path to shorten the signal acquisition time. The file can be generated by the utility fftw_wisdom in the Pocket SDR package.

Receiver Specific Options

Set all options default

The screenshot shows the 'System Options' dialog box with the following settings:

Option	Value
Epoch Interval for PVT (s)	1.0
Max Epoch Lag for PVT (s)	1.0
Elevation Mask for PVT (°)	15
Correlator Spacing (chip)	0.1
Integration Time for Acquisition (s)	0.02
Integration Time for DLL (s)	0.02
DLL Loop Filter Bandwidth (Hz)	0.5
PLL Loop Filter Bandwidth (Hz)	10.0
FLL Loop Filter Bandwidth Wide (Hz)	10.0
FLL Loop Filter Bandwidth Narrow (Hz)	5.0
Max Doppler Frequency to Search Signal (Hz)	5000
C/N0 Threshold for Signal Locked (dB-Hz)	34.0
C/N0 Threshold for Signal Lost (dB-Hz)	30.0
Carrier Lock Threshold (PLI)	0.25
Lost Decision Count (windows)	4
Bump Jump for BOC Modulation	OFF
Signal Acquisition Mode	Full-Search
Max Code Length for direct acquisition (ms)	4.0
FFTW Wisdom Path	C:/share/PocketSDR/python/fftw_wisdom.txt
Receiver Options	

Buttons at the bottom: Set Default, OK, Cancel.

Array Options Dialog

Array Element Horizontal Plot (Antenna Geometry)

Number of Array CH enabled.

Antenna Geometry Clear and Update Buttons

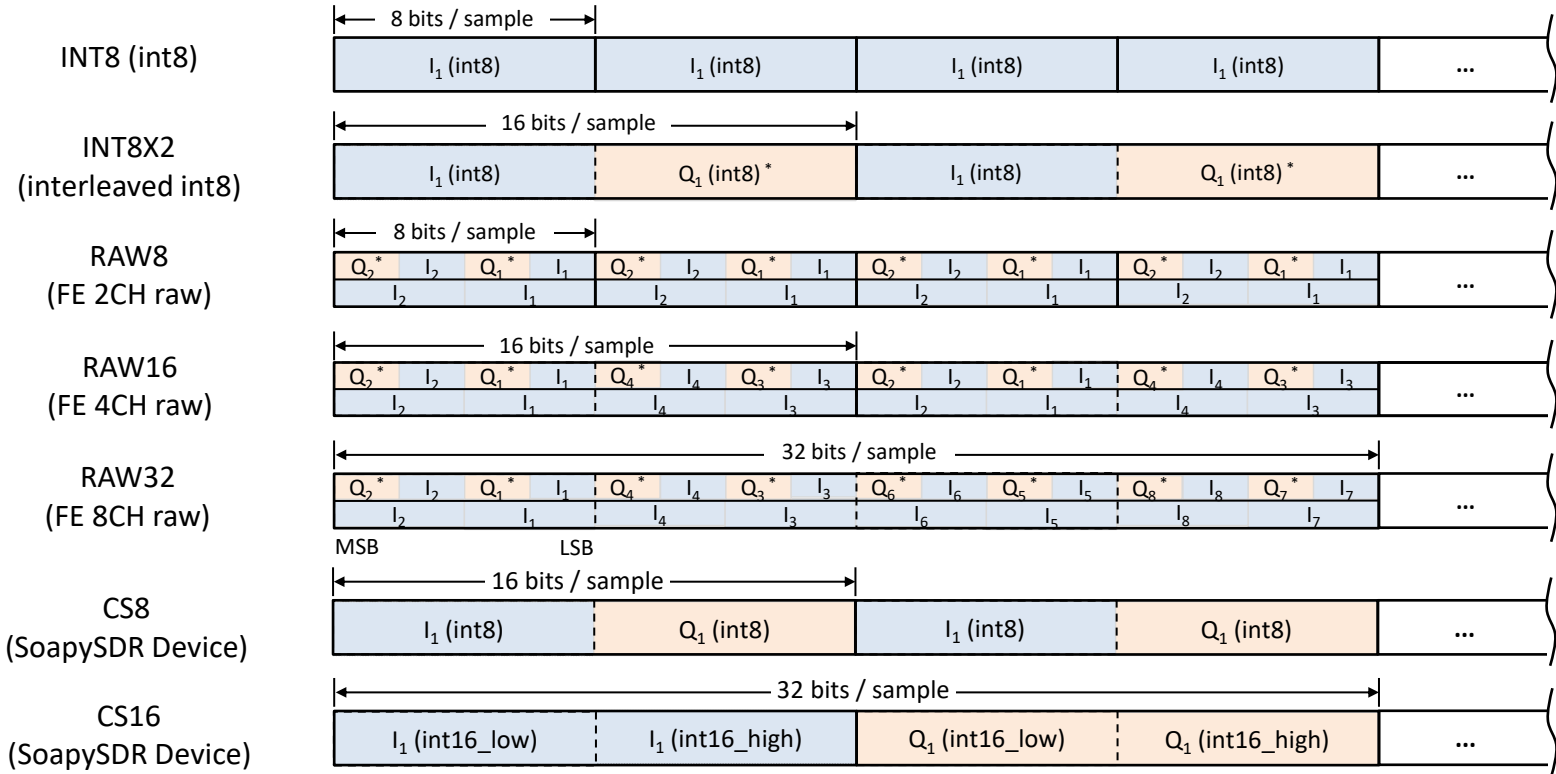
Antenna Geometry Load and Save Buttons

Enable/Disable and Positions X, Y, Z (m) in body-frame for each Antenna Element.

RF CH	POSX	POSY	POSZ
☒ CH1	0.0000	0.0000	0.0000 m
☒ CH2	0.0000	0.0951	0.0000 m
☒ CH3	0.0824	0.0476	0.0000 m
☒ CH4	0.0824	-0.0476	0.0000 m
☒ CH5	0.0000	-0.0951	0.0000 m
☒ CH6	-0.0824	-0.0476	0.0000 m
☒ CH7	-0.0824	0.0476	0.0000 m
☐ CH8	0.0000	0.0000	0.0000 m

IF Data File Formats

I_i : i-CH I-sample, Q_i : i-CH Q-sample * Q-polarity inverted due to MAX2771 convention



2 bits IQ-sampling (RAW8/16/32)

bits	00	01	10	11
I	+1	+3	-1	-3
Q*	-1	-3	+1	+3

2 bits I-sampling (RAW8/16/32)

bits	**00	**01	**10	**11
I	+1	+3	-1	-3

3 bits I-sampling (RAW8/16/32)

bits	0000	1000	0001	1001	0010	1010	0011	1011
I	+1	+3	+5	+7	-1	-3	-5	-7

IF Data Tag File (<path>.tag)

```

PROG = Pocket SDR
TIME = 2024/11/01 20:48:00.776   (Start Time GPST)
FMT = RAW16                       (Format)
F_S = 24                          (Sampling Freq. MHz)
F_L0 = 1568,1237.8,1176.45,1278.75 (LO Freq. MHz)
IQ = 2,2,2,2                     (Sampling, 1:I, 2:IQ)
BITS = 2,2,2,2                   (Sampling Bits)
    
```

Receiver Logs

\$TIME,time,year,mon,day,hour,min,sec,tsys (time info)
\$OBS,time,year,month,day,hour,min,sec,sat,code,cn0,pr,cp,dop,lli,fcn (OBS data)
\$NAV,time,sat,sig,nerr,size,data (raw NAV data)
\$POS,time,year,month,day,hour,min,sec,lat,lon,hgt,Q,ns,stdn,stde,stdu,dtr (position solution)
\$SAT,time,year,month,day,hour,min,sec,sat,pvt,obs,cn0,az,el,res (satellite info)
\$EPH,time,sat,sig,IODE,IODC,SVA,SVH,Toe,Toc,Ttr,A,e,i0,OMEGA0,omega,M0,delta-n,OMEGAdot,Idot,Crc,Crs,Cuc,Cus,Cic,Cis,Toes,Fit,Af0,Af1,Af2,TGD,code,flag (GPS/Galileo/QZSS/BeiDou/NavIC decoded ephemeris)
\$EPH,time,sat,sig,tb,fcn,SVH,SVA,age,Toe,Tof,pos-x,pos-y,pos-z,vel-x,vel-y,vel-z,acc-x,acc-y,acc-z,tau-n,gamma-n,delta-tau-n (GLONASS decoded ephemeris)
\$ALM,time,sat,sig,svh,week,A,e,i0,OMG0,omg,M0,OMGd,toas,f0,f1 (GPS,Galileo,QZSS,BeiDou decoded almanac)
\$ALM,time,sat,sig,svh,taun,lambda,di,eps,omg,tlambda,dT,dTd,frq (GLONASS decoded almanac)
\$CH,time,ch,rfch,sat,sig,prn,lock,cn0,coff,dop,adr,ssync,bsync,fsync,rev,srev,err_phas,err_code,tow_v,tow,week,type,nnav,nerr,nlol,nfec (receiver channel info)
\$LOG,time,message (error, warning, general info message)

\$TIME
 time receiver time (s)
 year year (2000-2099)
 mon month (1-12)
 day day (1-31)
 hour hour (0-23)
 min minute (0-59)
 sec second (0.000-59.999)
 tsys time system (GPST or UTC)

\$OBS
 time receiver time (s)
 year,month,day solution day (GPST)
 obs data day (GPST)
 hour,min,sec solution time (GPST)
 sat satellite ID
 code RINEX obs code
 cn0 C/N0 (dB-Hz)
 pr pseudorange (m)
 cp carrier phase (cyc)
 dop Doppler frequency (Hz)
 lli loss-of-lock indicator
 fcn frequency channel number for GLONASS FDMA signals

\$NAV
 time receiver time (s)
 sat satellite ID
 sig signal ID
 nerr number of corrected error
 bits
 size raw NAV data size (bits)
 data raw NAV data HEX dump
\$POS
 time receiver time (s)
 year,month,day solution day (GPST)
 hour,min,sec solution time (GPST)
 lat solution latitude (deg,+north,-south)
 lon solution longitude (deg,+east,-west)
 hgt solution ellipsoidal height (m)
 Q quality flag (=5: single)
 ns number of valid satellites
 stdn solution standard-dev north (m)
 stde solution standard-dev east (m)
 stdu solution standard-dev up (m)
 dtr receiver clock bias (s)

\$SAT
 time receiver time (s)
 year,month,day solution day (GPST)
 hour,min,sec solution time (GPST)
 sat satellite ID
 pvt PVT status (0:not used,1:used)
 obs L1 obs data status (0:unavailable,1:available)
 cn0 L1 signal C/N0 (dB-Hz)
 az azimuth angle (deg)
 el elevation angle (deg)
 res L1 pseudorange residual (m)

\$EPH
 time receiver time (s)
 sat satellite ID
 sig signal ID
 ... decoded ephemeris parameters

\$ALM
 time receiver time (s)
 sat satellite ID
 sig signal ID
 ... decoded almanac parameters

\$LOG
 time receiver time (s)
 message error, warning or general info message

\$CH
 time receiver time (s)
 ch receiver channel number
 rfch RF channel number
 sat satellite ID
 sig signal ID
 prn PRN number
 lock lock time (s)
 cn0 C/N0 (dB-Hz)
 coff code offset (ms)
 dop Doppler frequency (Hz)
 adr accumulated Doppler range (cyc)
 ssync secondary code sync flag (0:async,1:sync)
 bsync symbol/bit sync flag (0:async,1:sync)
 fsync frame sync flag (0:async,1:sync)
 rev primary code polarity (0:normal,1:reversed)
 srev secondary code polarity (0:normal,1:reversed)
 err_phas phase error (cyc)
 err_code code error (10^-6 s)
 tow_v tow valid flag (0:invalid,1:valid,2:ambiguity unresolved)
 tow time of week (ms)
 week week number (week)
 type navigation subframe or message type
 nnav navigation subframe/message count
 nerr error subframe/message count
 nlol loss-of-lock count
 nfec number of error corrected (bits)